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[Competitive Coding]

Q) Given three integers n, a and b , return the n^{th} magical no. Since the answer may be large, return mod of (10^9+7) .

A +ve integer is magical if number is divisible by either a or b

Sol) This question will

No. is magical if:

divisible by a || divisible by b

→ we need the multiples of a or b .

→ intuitive approach would be to start from 1 and keep checking each number, if divisible by a or b , consider it magical, increase count.

↳ when ~~count~~ count gets n , that particular num's mod with 10^9+7 will be returned.

Algorithm:

1) Initialise $\text{count} = 0$ and $\text{num} = 1$

2) ~~do~~ while ~~count~~ $(\text{count} \neq n)$;

→ if num is divisible by a or b , increase count by 1 and also increase num by 1

→ else increase num by 1

3) loop ends when $\text{count} = n$, ~~not needed~~ so, take mod of current value of num with 10^9+7

4) Return that value

Time Complexity = linear or $O(n)$